

NAME: _____ PERIOD: _____ DATE: _____

Narrower Is Not Automatically Better

AP Statistics · Unit 3 · Topic 3.3 · B: 2026-11-30 · E: 2027-01-04

[STAR] TODAY'S PROBLEM

A library system has 20,000 adult cardholders and wants to estimate p , the proportion who used a digital audiobook last month.

Plan A: A computer takes an SRS of 200 cardholder IDs. All answer; 116 say yes.

Plan B: A newsletter link lets cardholders choose to respond once. Of 800 respondents, 520 say yes.

Eli: “Use Plan B. Its larger sample gives the narrower interval.”

Dana: “Use Plan A. Entry into the sample matters more than the narrowest output.”

Plan counts

Plan	N	n	Yes	No
A	20,000	200	116	84
B	20,000	800	520	280

FIRST TAKE — before the video (5 min)

Which plan, if either, should estimate all adult cardholders? Cite one collection detail.

[BOOK] ONE-SAMPLE INTERVAL FOR A PROPORTION (keep on the page)

Use a one-sample z -interval for p : $\hat{p} \pm z^* \sqrt{\hat{p}(1-\hat{p})/n}$; for 95% confidence, $z^* = 1.96$. Check (1) random data collection, (2) when sampling without replacement, $n \leq 0.10N$, and (3) at least 10 observed successes and 10 failures. A smaller formula margin of error describes less random sampling variation; it does not repair systematic error caused by who chooses to respond.

Finish after the video:

DOK 1 (a) Define the population parameter p in context and name the confidence-interval procedure that would be appropriate when its conditions are met.

DOK 2 (b) For each plan, compute $\hat{p}$, check the random, 10 percent, and large-count conditions, and calculate the 95 percent formula interval. Mark any nondefensible population interval.

DOK 3 (c) Decide which plan can support a 95 percent interval for all cardholders. Cite recruitment and a numerical condition, compare interval widths, and explain the reader consequence of reporting Plan B's output.

Frame: I would report Plan ____ because _____, and the numerical check _____ is satisfied.

Frame: Although Plan B's formula width is _____, it would mislead readers because _____.

EXIT — turn this in

One thing the video changed about my first take: _____